

Software Requirements Specification (SRS)

Project: Blood Bank Management System

Version: 1.0

Authors: <Instructor / Student>

Date: 08-09-2026

Status: Model Answer / Proposed

Revision history

Version	Date	Author	Change summary
1.0	08-09-2026	Student	Initial SRS for Blood Bank Management System

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Blood Bank Management System (BBMS). It defines the functional and non-functional requirements, interfaces, security requirements, verification criteria, and traceability needed to develop and evaluate a system for managing blood donors, blood collection, blood inventory, blood requests, issue/dispatch, and administrative activities.

1.2 Scope

The system covers donor registration and eligibility records, blood donation records, blood-group and component inventory, hospital/patient blood requests, request approval, blood unit issue, stock monitoring, expiry tracking, notifications, reporting, and administrator management. It excludes laboratory equipment control, hospital internal clinical systems, and external payment processing beyond interfaces required by the BBMS.

1.3 Audience

Developers, database designers, QA engineers, system administrators, blood-bank staff, hospital/requesting staff, course instructors, and assessment evaluators.

1.4 Definitions

- BBMS – Blood Bank Management System
- Donor – A registered person who donates blood.
- Recipient/Patient – A person for whom blood or a blood component is requested.
- Blood Unit – A collected and uniquely identified unit of blood or blood component.
- Component – A separated blood product such as whole blood, packed red cells, plasma, or platelets.
- FIFO/FEFO – Inventory strategies; FEFO means First-Expire, First-Out.
- RBAC – Role-Based Access Control.
- OTP – One-Time Password.
- API – Application Programming Interface.
- Audit Log – Tamper-resistant record of important system actions.

2. Overall description

2.1 Product perspective

The BBMS is a centralized information system consisting of a web-based user interface, application services, a relational database, authentication and authorization services, notification services, and reporting facilities. It provides controlled access to blood-bank data and maintains traceability from donor registration and collection through inventory and issue.

2.2 Major product functions (detailed)

- Register and manage donor profiles.
- Record donor eligibility and donation history.
- Register blood collection and assign unique blood-unit identifiers.
- Record blood group and blood component information.
- Maintain real-time blood inventory and stock levels.
- Track blood-unit expiry dates and availability status.
- Register hospitals and authorized requesters.
- Create and manage patient/blood requests.
- Approve, reject, prioritize, and fulfill blood requests.
- Issue blood units using appropriate inventory-selection rules.
- Prevent allocation of expired, quarantined, or unavailable units.
- Generate inventory, donation, request, issue, and expiry reports.
- Send alerts for low stock, upcoming expiry, and request status.
- Manage users, roles, permissions, and audit logs.
- Support backup/recovery and administrative monitoring.

2.3 User roles and characteristics

Role	Characteristics
Donor	Provides registration and donation information; basic

	computer literacy expected.
Blood Bank Staff	Registers donors, records collections, manages inventory, and fulfills requests.
Hospital/Requester Administrator	Submits and tracks authorized blood requests.
Administrator	Manages users, roles, configuration, reports, and audit activities.
System/Database Administrator	Maintains deployment, database, backups, and system health.

2.4 Operating environment

Client: modern web browser on desktop/laptop/tablet. Server: application server capable of hosting the BBMS. Database: relational DBMS. Network: secure TCP/IP network or Internet connection. The system should support common institutional Windows/Linux environments and standard browser-based access.

2.5 Constraints

- Blood-unit records must be uniquely identifiable and traceable.
- Access to donor, patient, and inventory information must be role-controlled.
- The system must preserve inventory consistency when units are reserved or issued.
- Only authorized staff may change critical blood-unit status or issue records.
- The implementation must use a relational database with integrity constraints for core entities.

3. External interface requirements

3.1 User interfaces

The primary interface shall be a responsive web application with dashboards for each role. Forms shall provide validation for mandatory fields, clear status messages, searchable tables, confirmation before critical actions, and accessible navigation.

3.2 Hardware interfaces

- Desktop/laptop/tablet used by authorized users.
- Optional barcode/QR scanner for blood-unit identification.
- Optional printer for blood-unit labels and reports.

3.3 Software interfaces

- Relational database management system for persistent storage.
- Authentication/identity service for login and role verification.
- Email/SMS/notification service for alerts where configured.
- Optional barcode/QR service for blood-unit identification.

3.4 Communications

All client-server communication shall use HTTPS/TLS. API requests shall use authenticated sessions or tokens. Failed requests shall return clear error responses without exposing sensitive information.

Source/template structure adapted from the uploaded ATM SRS template, including its sections, requirement tables, acceptance criteria, security objectives/requirements, UML diagrams, and RTM. NfileciteÖturn0file0ÖL20-L27Ö

4. System features (detailed)

Each functional requirement is uniquely identified as BBMS-F-####. Requirements use 'shall' statements and include priority, stakeholder/source, acceptance criteria, and test-case references.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria	Test case ref
BBMS-F-001	The system shall allow authorized staff to register a donor with a unique donor ID and required personal/contact details.	Functional	High	Blood Bank Staff	A valid donor record is created and assigned a unique ID.	TC-Donor-01
BBMS-F-002	The system shall record donor eligibility information and shall prevent donation registration when configured eligibility rules are not satisfied.	Functional	High	Blood Bank Staff	Ineligible donor cannot proceed to collection registration.	TC-Donor-02
BBMS-F-003	The system shall maintain donor donation history including donation date, blood group, collection record, and status.	Functional	High	Blood Bank Staff	Completed donation appears in the donor history.	TC-Donor-03
BBMS-F-004	The system shall create a unique blood-unit ID for every accepted blood collection.	Functional	High	Blood Bank Staff	No two active blood units have the same identifier.	TC-Collection-01
BBMS-F-005	The system shall record blood group, component type, collection date, expiry date, and status for each blood unit.	Functional	High	Blood Bank Staff	A complete unit record can be stored and retrieved.	TC-Collection-02
BBMS-F-006	The system shall maintain current inventory counts by blood group, component, and availability status.	Functional	High	Blood Bank Staff	Inventory totals change correctly after collection, reservation, and issue.	TC-Inv-01
BBMS-F-007	The system shall automatically mark a blood unit as expired when its configured expiry date is reached and shall prevent its issue.	Functional	High	System/Staff	Expired unit is unavailable for allocation.	TC-Inv-02
BBMS-F-008	The system shall allow authorized staff to quarantine, release, or discard a blood unit with a recorded reason and audit entry.	Functional	High	Blood Bank Staff	Status change requires authorization and reason; audit entry is created.	TC-Inv-03
BBMS-F-009	The system shall allow authorized hospitals/requesters to create blood requests specifying patient/request details, blood group, component, quantity, priority, and required date.	Functional	High	Hospital/ Requester	Valid request receives a unique request ID and pending status.	TC-Req-01
BBMS-F-010	The system shall validate requested blood group/component and reject	Functional	High	Hospital/ Requester	Invalid request is rejected with field-level validation.	TC-Req-02

	incomplete or invalid request data.					
BBMS-F-011	The system shall allow authorized blood-bank staff to approve, reject, or place a request on hold and record the decision.	Functional	High	Blood Bank Staff	Request status and decision history are updated correctly.	TC-Req-03
BBMS-F-012	The system shall allocate available eligible blood units to an approved request using FEFO where applicable.	Functional	High	Blood Bank Staff	Allocation selects an eligible available unit and records the association.	TC-Alloc-01
BBMS-F-013	The system shall prevent issue of blood units that are expired, quarantined, already issued, or unavailable.	Functional	High	Blood Bank Staff	System blocks invalid issue and displays a clear reason.	TC-Alloc-02
BBMS-F-014	The system shall record every blood-unit issue with unit ID, request ID, requester, date/time, quantity, and issuing staff member.	Functional	High	Blood Bank Staff	Issue transaction is stored and traceable end-to-end.	TC-Issue-01
BBMS-F-015	The system shall update inventory atomically after blood collection, allocation, issue, discard, or status changes.	Functional	High	System	No partial inventory update occurs when a transaction fails.	TC-Inv-04
BBMS-F-016	The system shall provide searchable and filterable records for donors, blood units, requests, and issue transactions to authorized users.	Functional	Medium	Staff/Admin	Authorized user can locate records using supported filters.	TC-Search-01
BBMS-F-017	The system shall generate reports for inventory levels, donations, issued units, pending requests, expired units, and discarded units.	Functional	Medium	Administrator/ Staff	Selected report is generated with correct records for the selected period.	TC-Report-01
BBMS-F-018	The system shall notify authorized users about low-stock conditions, upcoming expiry, and important request-status changes when notifications are enabled.	Functional	Medium	Staff/Admin	Configured event generates the appropriate notification.	TC-Notif-01
BBMS-F-019	The system shall allow administrators to create, deactivate, and assign roles to system users.	Functional	High	Administrator	Only administrators can modify user roles and deactivation status.	TC-Admin-01
BBMS-F-020	The system shall record audit events for	Functional	High	Administrator	Audit log contains actor, timestamp, action, and	TC-Audit-01

	authentication, critical inventory changes, approvals, issues, user/role changes, and other configured security events.				relevant record reference.	
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5. Non-functional requirements (detailed)

Req ID	Requirement	Category	Priority	Acceptance/Test
BBMS-NF-001	The system shall return normal search, form submission, and dashboard requests within 3 seconds for at least 95% of requests under normal expected load.	Performance	High	TC-Perf-01
BBMS-NF-002	The system shall provide at least 99.5% monthly availability excluding planned maintenance announced in advance.	Reliability	High	TC-Reliability-01
BBMS-NF-003	The system shall encrypt sensitive data in transit using TLS and protect sensitive stored data using appropriate database/application security controls.	Security	High	TC-Sec-01
BBMS-NF-004	The system shall maintain timestamped audit logs for critical actions and retain them for at least 1 year or the institution's configured retention period.	Audit/Data Retention	High	TC-Audit-02
BBMS-NF-005	The system shall provide a usable responsive interface with clear validation, consistent navigation, and accessibility support appropriate to the deployment environment.	Usability/Accessibility	Medium	TC-UX-01

5.1 Security

5.1.1 Security Objectives

- Confidentiality: protect donor, patient, requester, authentication, and operational information from unauthorized access or disclosure.
- Integrity and accountability: ensure blood-unit inventory and issue records cannot be altered or deleted without authorization and traceable audit evidence.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance/Test
BBMS-SR-001	The system shall enforce role-based access control so users can access only functions and records permitted for their role.	Security	High	TC-Sec-02
BBMS-SR-002	The system shall require secure authentication and shall terminate or	Security	High	TC-Sec-03

	expire inactive sessions according to configured policy.			
BBMS-SR-003	The system shall protect passwords using a strong one-way password hashing mechanism and shall never store plaintext passwords.	Security	High	TC-Sec-04
BBMS-SR-004	The system shall validate and sanitize user input and use parameterized database operations to prevent injection and related input-based attacks.	Security	High	TC-Sec-05
BBMS-SR-005	The system shall create audit records for security-sensitive actions, including failed logins, role changes, blood-unit status changes, approvals, and issue transactions.	Security	High	TC-Sec-06

6. Quality attributes & Acceptance tests

- Correctness: all high-priority functional requirements must be implemented and verified.
- Reliability: collection, allocation, and issue transactions must preserve database and inventory consistency.
- Security: all security requirements must pass their associated security tests.
- Usability: representative staff users shall be able to complete donor registration, inventory lookup, request processing, and issue workflows without ambiguous system states.
- Acceptance exit criteria: all high-priority requirements pass, no critical security defect remains open, inventory reconciliation shows no unexplained discrepancies, and the RTM has test coverage for all requirements.

Acceptance test suites

- Donor & Collection Tests
- Inventory & Expiry Tests
- Request & Allocation Tests
- Issue & Reporting Tests
- Performance Tests
- Security & Access-Control Tests
- Usability Tests

7. System models and diagrams

7.1 UML Use-Case Diagram 1 – Donor and Inventory Management

Actors: Blood Bank Staff, Administrator. Main use cases: Login, Register Donor, Check Eligibility, Record Donation, Create Blood Unit, View Inventory, Update Blood Unit Status, Track Expiry, Generate Inventory Report, Manage Users.

[Insert UML Use-Case Diagram here: Donor/Collection/Inventory workflow]

7.2 UML Use-Case Diagram 2 – Blood Request and Issue Management

Actors: Hospital/Requester, Blood Bank Staff, Administrator. Main use cases: Login, Create Blood Request, View Request Status, Approve/Reject Request, Allocate Blood Unit, Issue Blood Unit, Update Inventory, Generate Issue Report, Receive Notifications.

[Insert UML Use-Case Diagram here: Request/Approval/Allocation/Issue workflow]

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)
BBMS-F-001	Register donor	4 / DS-Donor-01	Donor Module	TC-Donor-01	N
BBMS-F-004	Create unique blood unit	4 / DS-Collection-01	Collection Module	TC-Collection-01	N
BBMS-F-006	Maintain inventory	4 / DS-Inventory-01	Inventory Module	TC-Inv-01	N
BBMS-F-009	Create blood request	4 / DS-Request-01	Request Module	TC-Req-01	N
BBMS-F-012	FEFO allocation	4 / DS-Allocation-01	Allocation Module	TC-Alloc-01	N
BBMS-F-014	Record blood issue	4 / DS-Issue-01	Issue Module	TC-Issue-01	N
BBMS-F-017	Generate reports	4 / DS-Report-01	Reporting Module	TC-Report-01	N
BBMS-F-019	Manage users/roles	4 / DS-Admin-01	Admin Module	TC-Admin-01	N
BBMS-F-020	Audit events	4 / DS-Audit-01	Audit Module	TC-Audit-01	N
BBMS-NF-001	Response time	5 / DS-Perf-01	WebUI/API	TC-Perf-01	N
BBMS-NF-002	Availability	5 / DS-Reliability-01	Deployment	TC-Reliability-01	N
BBMS-SR-001	RBAC	5.1.2 / DS-Sec-01	Auth Module	TC-Sec-02	N
BBMS-SR-003	Password protection	5.1.2 / DS-Sec-02	Auth/DB	TC-Sec-04	N
BBMS-SR-005	Security audit logging	5.1.2 / DS-Sec-03	Audit Module	TC-Sec-06	N